University of Oulu | Big Data Processing and Applications

Exploratory and Sentiment Analysis of Yelp Reviews using Apache Spark

Group 12
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8.5 GB
Dataset Size

5
JSON Tables

What We Will Cover

01 **Dataset & Motivation**

Why Yelp? Research questions, project scope

02 **Data Description**

5 tables, 8.5 GB, schema & statistics

03 **Exploratory Data Analysis**

Geography, categories, time, words

04 **Sentiment ML Pipeline**

Features, 3 classifiers, evaluation

05 **Business Reputation Analysis**

Scoring 6.99M reviews, closure prediction

06 **Results & Conclusions**

Comparison with related work, future work

Related Work

Industry and Academic Journals

* Roberto Sanchez. Sentiment analysis. https://github.com/rsanchez-dv/Sentiment_Analysis.git, 2021.

* Akhil Bharat Sisal. Title. https://norma.ncirl.ie/7617/1/akhilbharatsisal. pdf, 2023. Accessed: 2026-04-30.

* Mujiono Sadikin and Abi Fauzan. Evaluation of machine learning approach for sentiment analysis using yelp dataset. European Journal of Electrical Engineering and Computer Science, 7(6):58–64, 2023.

* Rachid Belaroussi, Sié Cyriac Noufe, Francis Dupin, and Pierre-Olivier Vandanjon. Polarity of Yelp Reviews: A Bert–LSTM Comparative Study. Big Data and Cognitive Computing, 9(5):140, 2025.

* SonyaZhang, LindaLy, NormanMach, and ChristianAmaya. Topic modelling and sentiment analysis of Yelp restaurant reviews. International Journal of Information Systems in the Service Sector (IJISSS), 14(1):1–16, 2022.

* Bhanu Prakash Reddy Guda, Mashrin Srivastava, and Deep Karkhanis. Sentiment analysis: Predicting Yelp scores. arXiv preprint arXiv:2201.07999, 2022.

Motivation & Research Questions

Why This Project?

* Businesses cannot manually inspect millions of reviews — automation is essential.
* Star ratings lack aspect-level cues: food quality, service speed, and ambience.
* 8.5 GB multi-table dataset — ideal proving ground for the full Spark ecosystem.

RQ1

What geographic, temporal, and categorical patterns characterise Yelp businesses and reviews?

RQ2

Can ML models trained on text + metadata accurately classify Negative / Neutral / Positive sentiment?

RQ3

Does model-derived sentiment correlate with real-world business closure?

The Yelp Open Dataset — 8.5 GB Across Five Tables

Review

6.99M rows

stars · text · date · useful / funny / cool

Business

150K rows

categories · location · is_open · stars

User

2.0M rows

review_count · average_stars · fans · elite

Checkin

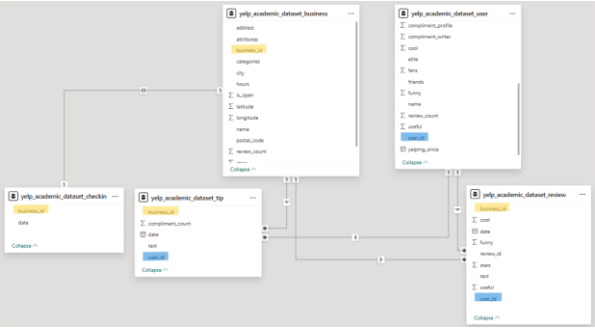
131K rows

comma-delimited timestamps per business

Tip

909K rows

short user tips · compliment_count



Relationships: Business ←→ Review (1:Many) · User ←→ Review (1:Many) · Business ←→ Checkin (1:1) · User ←→ Tip (1:Many)

Data Preprocessing — Two Pipelines, One Goal

Shared steps (both pipelines): Parse dates to TimestampType · Fix −1 values in useful/funny/cool → 0 · Drop null categories · Impute missing address from city + postal_code

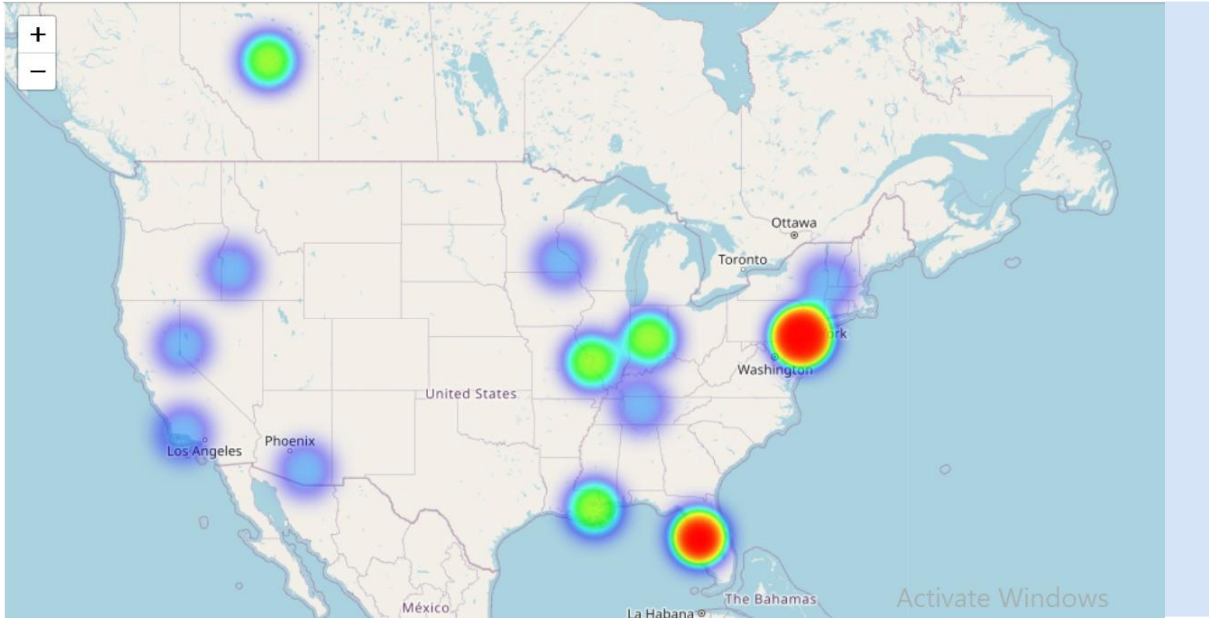
EDA Pipeline

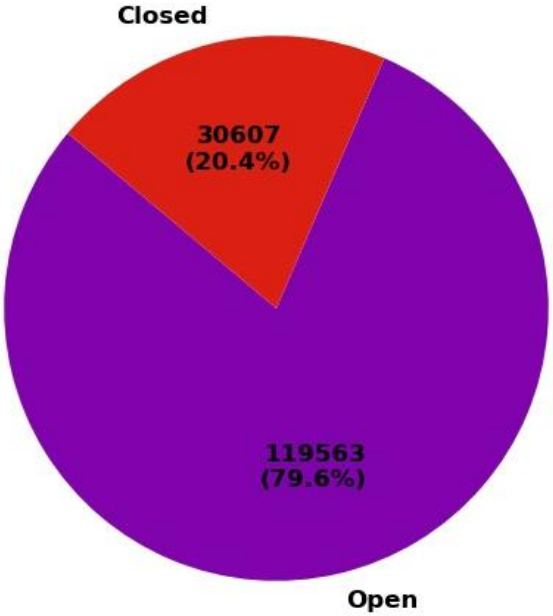
- Drop nested attributes & hours structs (>30% missing)
- Hand-craft 10 macro-categories from 200+ Yelp tags using Spark when/contains
- Resolves: 'Restaurants, Pizza' ≠ 'Pizza, Restaurants'
- Powers all geographic and category-level EDA analyses

ML Pipeline

- 5-table left-join → 6.99M analytical frame
- Nulls from joins → 0 or empty array
- Engineer: review_length, num_exclamations, num_caps
- Composites: user_activity = review_count + fans; user_engagement = useful + funny + cool
- Label: ≤2★ → Negative (0) · 3★ → Neutral (1) · ≥4★ → Positive (2)

EDA: Where Are the Businesses?



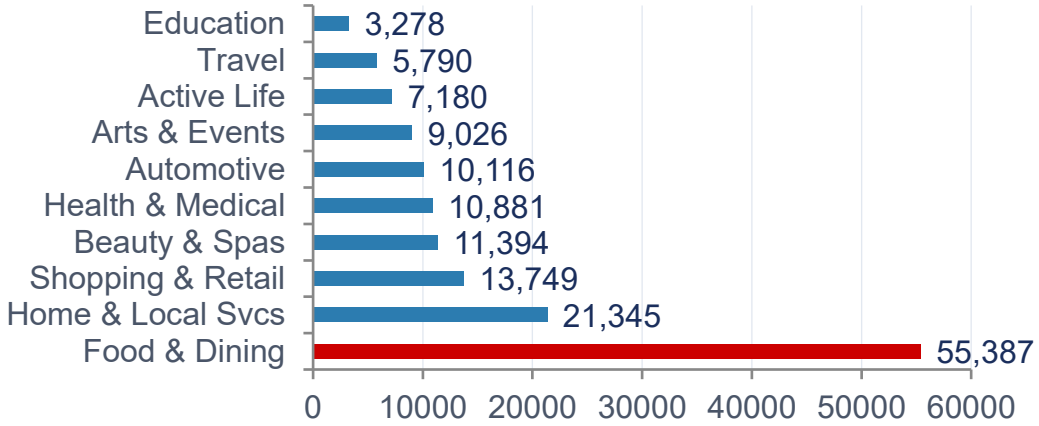


14,552	9,039	6,965	6,203
Philadelphia	Tampa	Nashville	New Orleans

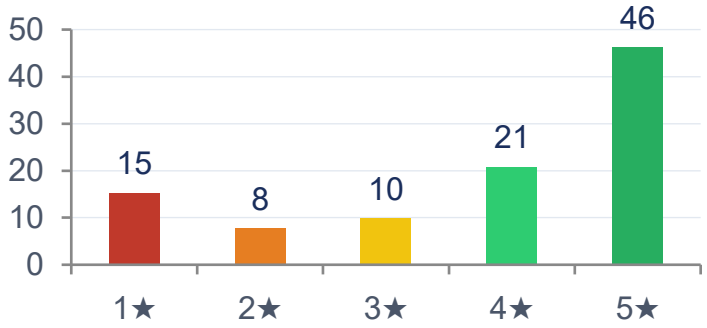
Dataset covers 11 US metro areas + Edmonton, Canada — geographically diverse but North American-centric (a known limitation).

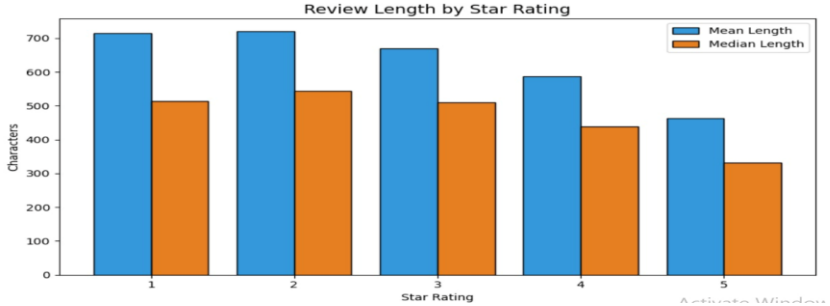
EDA: Business Categories and Star Ratings

Top 10 Business Categories



Star Rating Distribution (%)



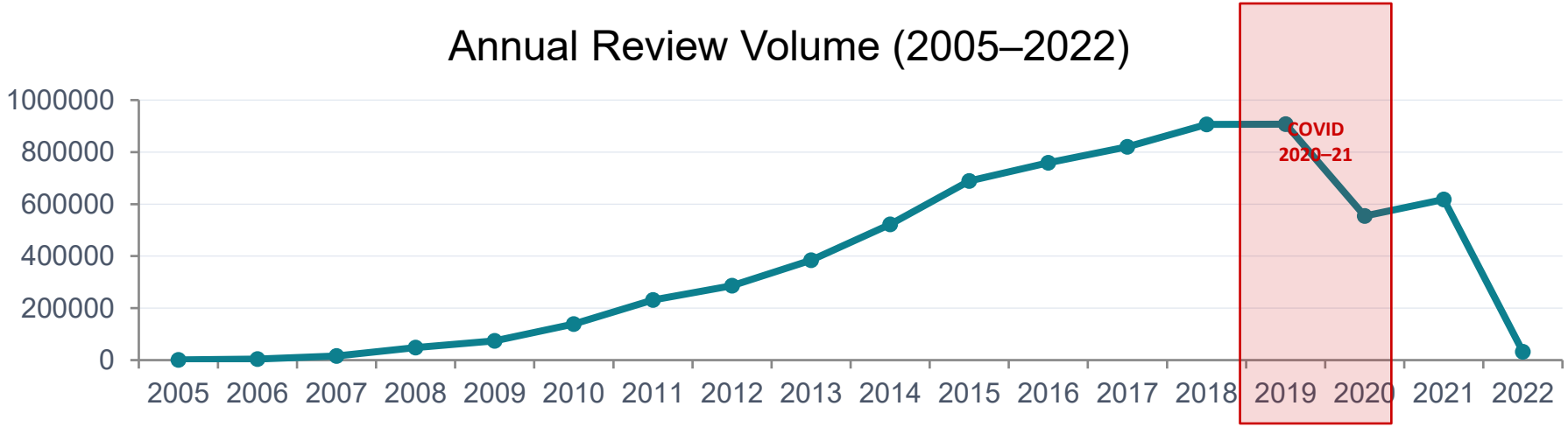


⚠ Class Imbalance

72% of reviews are 4–5★
Neutral (3★) = only 9.9%
→ Stratified undersampling required

Food & Dining dominates: 55,387 businesses (37% of total). Negative reviews average 150 chars longer than positive — dissatisfied customers write more.

EDA: When Do People Write Reviews?



Monthly Peak

July – August

Summer leisure dining drives the annual review surge

Evening Peak

18:00 – 21:00

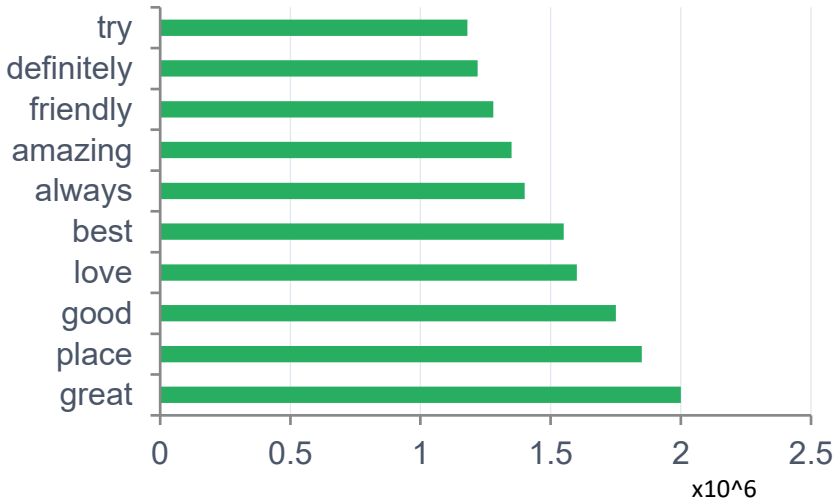
Most reviews written after dinner hours

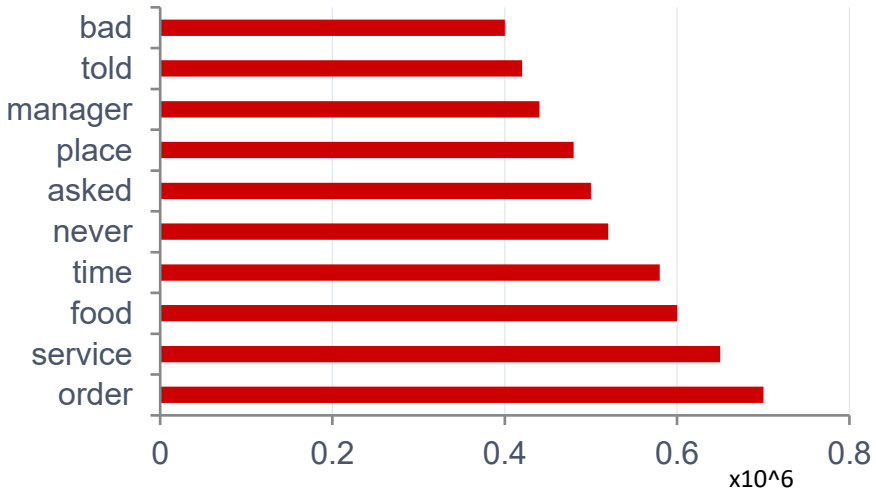
COVID Drop

–39% in 2020

From 907K (2019) to 554K — services disrupted

EDA: What Are Customers Saying?





Category sentiment: Active Life & Fitness and Arts & Entertainment show the highest positive ratios. Automotive and Home & Local Services have the highest negative ratios — consistent with Zhang et al. (2022), who found service quality and price to be the most sentiment-charged topics.

Machine Learning Pipeline — From Reviews to Sentiment

Raw Reviews 6.99M → Tokenise + Stop-words → CountVec TF-IDF vocab 2000 → Category Vectoriser vocab 200 → VectorAssembler + StandardScaler → LR / RF / MLP

Logistic Regression

Linear baseline — interpretable coefficient analysis per sentiment class

Params: λ=0.01 · 100 iterations · L2 regularisation · class weights

Train: 415K reviews

Random Forest

Captures non-linear interactions; provides feature importance rankings

Params: 50 trees · depth 15 · min 10 instances/leaf · √ feature subset

Train: 415K reviews

MLP (Neural Net)

Shallow neural baseline; bridges to LSTM architectures in related work

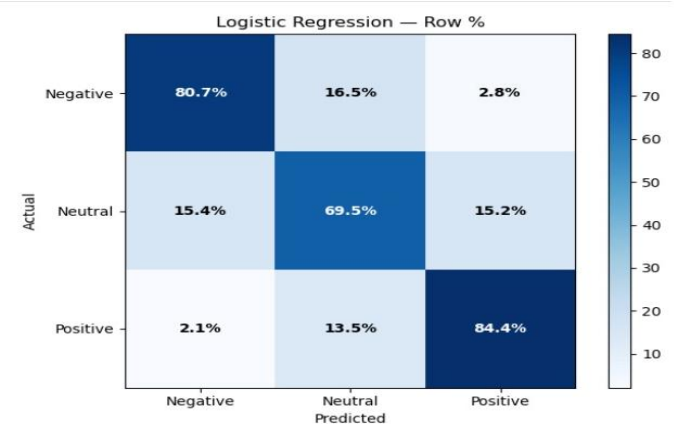
Params: Architecture [input, 64, 3] · L-BFGS solver · block 512

Train: 104K reviews*

** MLP uses 10% of featurised training set due to Rahti memory constraints. Stratified undersampling balances all three classes before splitting.*

ML Results — Three-Class Sentiment Classification

Model	Accuracy	F1 (weighted)	Neg Recall	Neutral Recall	Pos Recall
Logistic Regression	78.2%	78.2%	78.1%	66.3%	81.1%
Random Forest	69.5%	69.5%	73.3%	60.5%	69.0%
MLP (64 hidden)	72.7%	72.8%	~78%	~65%	~75%

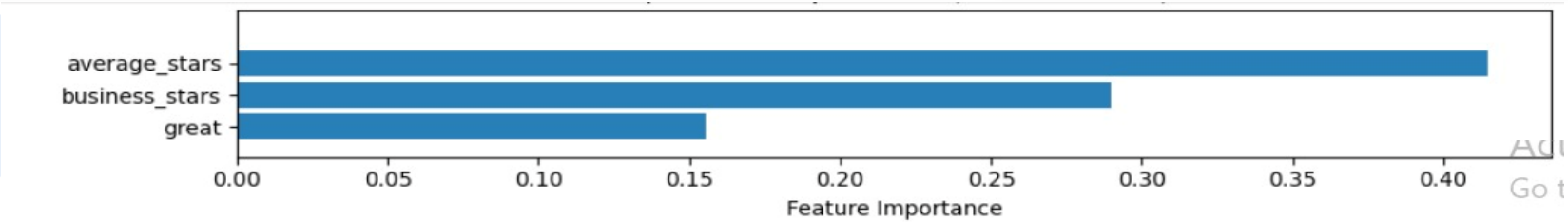


⚠ **Neutral class is the hardest to classify across all models.**

Three-star reviews contain genuinely mixed language — consistent with Belaroussi et al. (2025) who report the same difficulty across LSTM, Bi-LSTM, and all BERT variants.

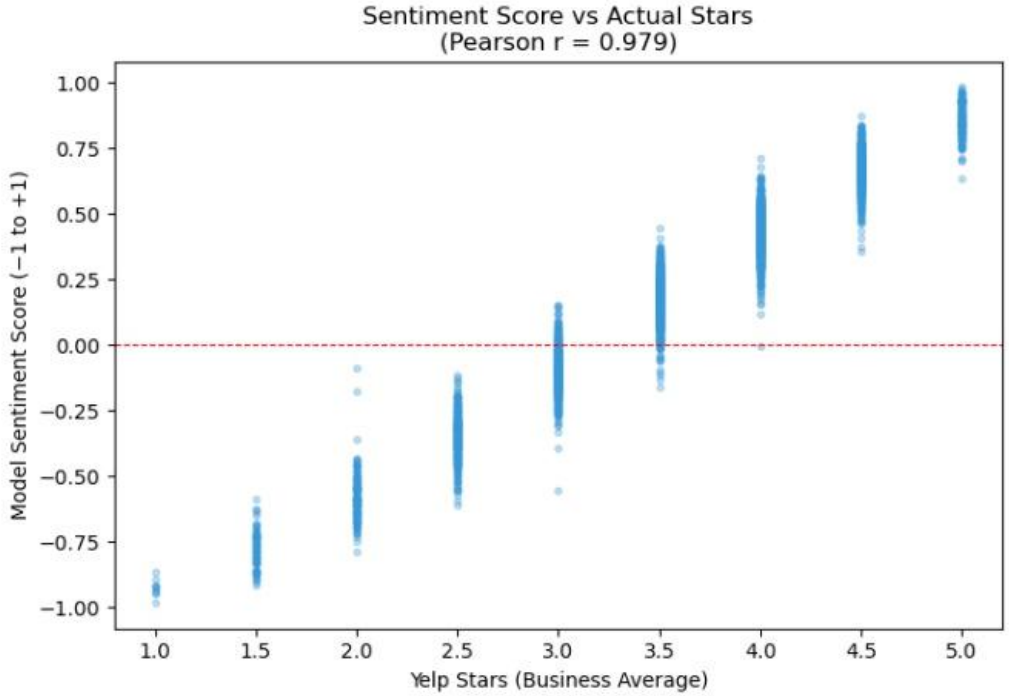
vs. Belaroussi et al. (2025):

Best LSTM → 77% on 107K balanced reviews
Our LR → 78.2% on 415K reviews
Comparable — without LSTM architecture



What the Model Learned — Feature Importance

Sentiment vs. Actual Stars (r = 0.979)



POSITIVE sentiment drivers

✓ amazing
✓ delicious
✓ fantastic
✓ great
✓ perfect
✓ love
✓ num_exclamations

NEGATIVE sentiment drivers

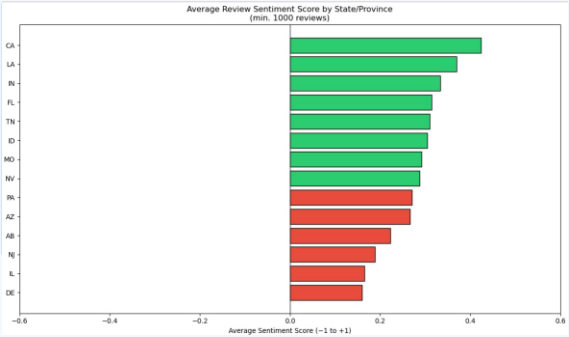
✗ worst
✗ horrible
✗ rude
✗ terrible
✗ waste
✗ disappointed
✗ bland

Guda et al. (2022) reported r ≈ 0.52 for the AFINN lexicon. Our model-derived score shows even stronger alignment — validating model quality.

Beyond Classification — Business Reputation Scoring

Reputation Score = P(Positive) – P(Negative) ∈ [−1, +1] applied to all 6.99M reviews

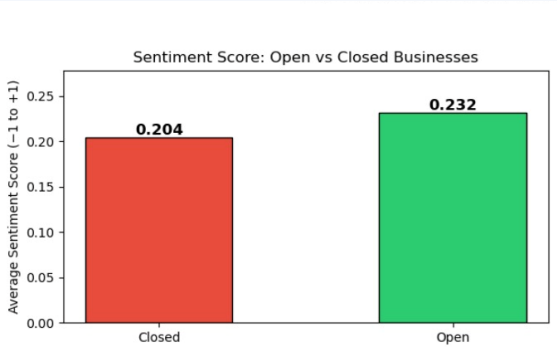
Average Sentiment Score by State



Min. 1K reviews Businesses

High positive sentiment in some states shows that service quality and price are the most sentiment-charged topics

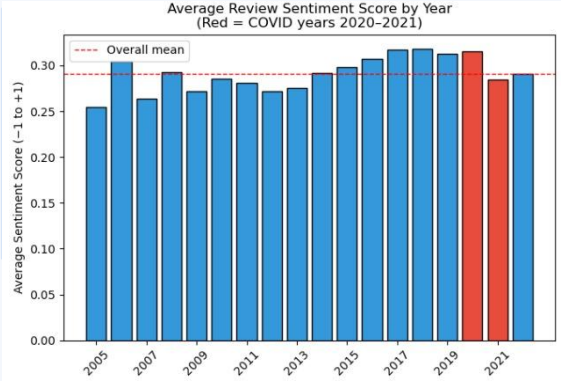
Open vs. Closed Businesses



Open: 0.232 vs Closed: 0.204

Sustained negative sentiment may predict business closure — a novel downstream finding not present in any related work.

Temporal Sentiment (2005–2022)



COVID dip 2020–2021

Sentiment scores dip in 2021, corroborating the COVID-19 service-quality deterioration implied by EDA review volume drop.

How Do Our Results Compare?

Aspect	Related Work	Our Work
Dataset scale	36K – 200K reviews	**6.99M reviews**
Processing platform	Single-node Python / Keras	**Apache Spark — distributed**
Feature approach	Text-only (TF-IDF or embeddings)	**TF-IDF + 15 engineered meta-features**
Best 3-class accuracy	77% (LSTM, Belaroussi 2025)	**78.2% (LR)**
Neutral class recall	68–70% (best LSTM)	66.3% (LR)
Class balancing	Balanced subset	**Stratified undersampling of full 7M**
Downstream validation	None in cited works	**Business reputation + closure prediction**

Our primary contribution is operating a complete EDA + ML pipeline at big-data scale — with interpretable, actionable business intelligence output.

Conclusions & Future Work

Key Findings

✓ Food & Dining dominates Yelp; 72% of reviews are 4–5★ — severe positive skew

✓ Negative reviews are ~150 char longer — dissatisfied customers write more

✓ LR (78.2%) matches BERT-era LSTM (77%) at 65× more training data

✓ Business reputation scores correlate strongly with star ratings (r = 0.979)

✓ Closed businesses show lower historical sentiment — ML signal closure risk

✓ Sentiment dipped in 2020–2021, confirming COVID-19 service disruption

✓ All material related to the project can be found here:
https:// github.com/FaizanTanwir/BDPA2026_Group-12_Project-YELP.git

Future Work

→ Fine-tuned BERT for improved neutral-class recall

→ Aspect-based sentiment (food / service / price) per Zhang et al. (2022)

→ Spark Structured Streaming for real-time reputation monitoring

→ Recency-weighted reputation scores